MILITARY SPECIFICATION

SEAT CUSHION INSERT, POLYURETHANE FOAM, PLASTIC, GENERAL SPECIFICATION FOR

This specification is approved for use by the Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

* 1.1 Scope. This specification covers the general requirements for plastic, open-celled, prefoamed, uncored, polyether polyurethane seat cushion inserts.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

* 2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-601	Boxes, Wood, Cleated-Plywood
PPP-B-636	Box, Shipping, Fiberboard

MILITARY

MIL-P-116	Preservation, Methods of

STANDARDS

MILITARY

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-143	Standards and Specifications, Order of Precedence for the Selection of
MIL-STD-794	Part and Equipment, Procedures for Packaging and Packing

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: the Engineering Division, San Antonio ALC/MMEDO, Kelly AFB, Texas 78241 by using the self addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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(Copies of specifications and standards required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents form a part of this specifications to the extent specified herein.

NATIONAL BUREAU OF STANDARDS

H4-1 - Federal Supply Code for Manufacturers.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D.C. 20402.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 3951-82 - Packaging, Commercial - 29 Jan 82

(Application for copies of ASTM publications should be addressed to the American Soceity for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

2.1.3 Order of precedence. In the event of a conflict between the text of this specification and this the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 First article. This specification makes provisions for first article testing.

3.2 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.3 Materials. The materials used in the manufacture of polyurethane foam for seat cushion inserts shall be high quality, uncored, open-celled, polyether polyurethane. The material shall be prefoamed and cut to the required shape.

3.4 Configuration. The seat cushion inserts shall have the dimensions and dimensional tolerances specified on the applicable drawings.

3.5 Performance.

3.5.1 Load deflection characteristics (50 square inch indenter foot). Deflection by means of a 50 square inch deflector foot shall produce the values specified in table I. The size of the test sample shall consist of the entire seat cushion insert. Samples less than 4 $\pm$ 1/8 inches in height shall be plied up, without the use of cement, to a height of 4 +1/8 inches. If the entire seat cushion insert has dimensions less than 15 by 15 inches or a height greater than 4 +1/8 inches, a test sample of the seat cushion insert material having dimensions of 18 by 18 by 4 +1/8 inches shall be submitted for tests.

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TABLE I. Load deflection characteristics requirements (50 square inch indenter foot).

Percent Deflection	Load
25	40 ± 5 pounds
50	60 ± 6 pounds
65	80 ± 8 pounds
75	130 ± 20 pounds
80	165 ± 40 pounds
85	180 pounds minimum

3.5.2 Density. The density of the polyurethane shall be 1.6 + 0.2 pounds/cubic feet.

3.5.3 Tensile strength. The tensile strength of the seat cushion insert material shall have a value of 12 pounds per square inch minimum.

3.5.4 Elongation. The value of the ultimate percent elongation for the insert material shall be 180 percent minimum.

3.5.5 Compression set. The percent loss of original height shall be not more than 15 percent.

3.5.6 Air oven aging. The allowable percent change in load deflection values after air oven aging shall be ± 10 percent maximum,

3.5.7 Humidity aging.

3.5.7.1 After humidity aging, the percent loss of original height shall be not more than 15 percent.

3.5.7.2 After humidity aging, the percent change in load deflection characteristics for 25, 50, 65, 75, 80, and 85 percent deflections shall be not more than 25 percent for the 50 square inch indenter foot test.

3.5.8 Tear resistance. The tear resistance of the insert material shall be not less than 1.5 pounds per inch for all specimens.

3.6 Identification of product. The following information shall be stenciled on each seat cushion insert in accordance with MIL-STD-130:

a. Title and number of this specification.

b. Contract number and date.

c. Stock number.

d. Manufacturer's name (if desired, the manufacturer's internal designations and trade name may be included).

e. Applicable drawing number.

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The identification marking ink shall be nontoxic, waterproof, fade resistant, and resistant to obliteration in normal usage. Letter and numeral sizes shall be 1/2 inch. The identification marking shall be on the top or bottom surface of the seat cushion insert, located at the discretion of the manufacturer.

3.7 Workmanship. The seat cushion inserts shall be constructed in a thoroughly workmanlike manner. They shall be clean and free from any defects that might affect their appearance or functionability.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests. The inspection and testing of seat cushion inserts shall be classified as follows:

a. First article tests (see 4.4)

b. Acceptance tests (see 4.5)

4.3 Test conditions.

4.3.1 Atmospheric conditions. Unless otherwise specified, all tests required by this specification shall be made at a temperature of 73° +2°F and a relative humidity of 50 +2 percent.

4.3.2 Test specimens. Test specimens shall consist of the entire seat cushion inserts or samples, as specified, which shall be cut from seat cushion inserts. No more than one test sample for each specified test shall be cut from the same seat cushion insert. Unless otherwise specified, height of the specimens shall be 4 $\pm$ 1/8 inches.

4.4 First article testing.

4.4.1 First article test samples. The contractor shall subject specimens to the first article tests specified in 4.4.3. Three seat cushion inserts shall be required for the load deflection characteristics tests.

4.4.2 First article test samples for the procuring activity. The contractor shall submit three seat cushion inserts to the procuring activity.

4.4.3 First article tests. First article tests shall consist of all the tests described under 4.6.

4.5 Acceptance tests. Acceptance tests shall consist of:

a. Individual tests.

b. Sampling plan and tests.

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4.5.1 Individual tests. Each seat cushion insert shall be subjected to the examination of product test as described under 4.6.

4.5.2 Sampling plan and tests.

4.5,2.1 Lot. A lot shall consist of seat cushion inserts manufactured under essentially the same conditions and submitted for acceptance at substantially the same time.

4.5.2.2 Sampling tests. Four seat cushion inserts shall be selected at random from each lot of one hundred inserts or fraction thereof and subjected to the load deflection characteristics tests described under 4.6.

4.5.2.2.1 Rejection and retest. When one or more items from a lot fail to meet the specification, acceptance of all items in the lot shall be withheld until the extent and cause of failure are determined. The contractor shall explain fully to the Government representative the cause of failure and the action taken to preclude recurrence. After corrections have been made, all necessary tests shall be repeated.

4.5.2 .2.2 Individual tests may continue. For production reasons, individual tests may be continued pending the investigation of a sampling test failure. Final acceptance of the entire lot or lots produced shall not be made until it is determined that all items meet all the requirements of the specification.

4.5.3 Defects in items already accepted. The investigation of a test failure could indicate that defects may exist in items already accepted. If so, the contractor shall fully advise the procuring activity of all the defects likely to be found and the method of correcting them.

4.6 Test methods.

4.6.1 Examination of product. The length, width, and height of seat cushion inserts shall be measured to the nearest 1/16 inch to determine conformance to the dimensions and tolerances specified on the applicable drawings. The seat cushion inserts shall be visually examined to determine conformance to the requirements of this specification with respect to materials, workmanship, and marking.

4.6.2 Load deflection characteristics. The following load deflection characteristics tests shall be conducted to determine compliance with the requirements of 3.5.1 and Table I.

4.6.2.1 Fifty square inch indentor foot. An apparatus having a flat circular indenter foot 50 square inches in area, connected by means of a swivel joint to a load measuring device and mounted in such a manner that the specimen

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can be deflected at a suitable rate, shall be used for this test. The apparatus shall be arranged to support the specimen on a level horizontal plate perforated with 0.25 inch holes on 0.75 inch centers to allow for rapid escape of air during the test. The seat cushion insert shall first be twice deflected to 75 +5 percent of its original height by using plates or rollers as a means of obtaining the deflection. The specimen shall then be allowed to rest a period of 10 +5 minutes. The specimen shall then be placed in position on the supporting plate of the apparatus and the indenter foot brought into contact with the specimen. The height shall be determined after applying a total preload of 1 pound. The specimen shall then be compressed 25 percent of this height and the final load observed after 1 minute. The same procedure shall be followed for 50, 65, 75, 80 and 85 percent load deflection characteristics.

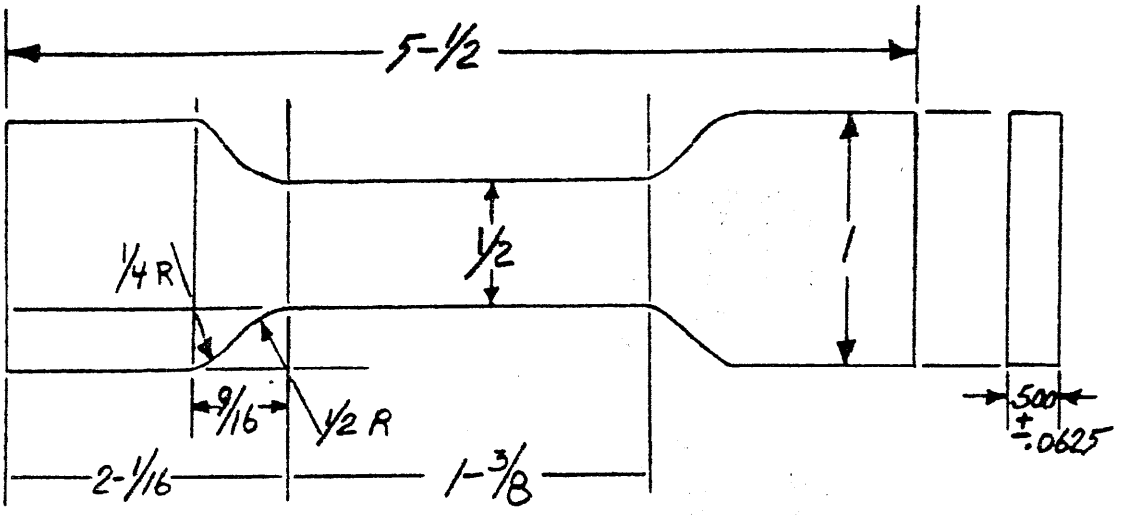
4.6.3 Density. Three representative specimens of regular shape shall be cut from three different seat cushion inserts and tested to determine compliance with the requirements of 3.5.2. The specimens shall be not less than 1 cubic inch in volume and shall be free from surface skin. The volume of the specimens shall be calculated and the specimens weighed on a balance or scale graduated to permit weighings within ± 1 percent of the weight to be measured. The density shall be reported in-pounds/cubic feet and shall be calculated as follows:

$$\text{Density in pounds/cubic feet} = \frac{\text{Specimen weight in pounds}}{\text{Specimen volume in cubic feet.}}$$

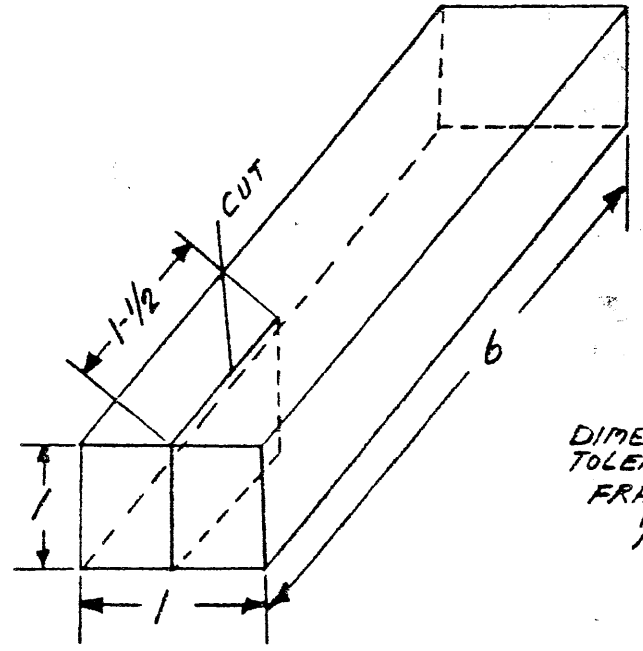
4.6.4 Tensile strength. Three test specimens as shown on figure 1 shall be out from three different seat cushion inserts and tested to determine compliance with the requirements of 3.5.3. The specimens shall be free from nicks and ragged edges. The tensile test shall be made on a power-driven machine equipped with a dial or scale indicator that will remain at the point of maximum load after rupture. The rate of travel of the power-driven grips shall be 20 ± l inches per minute and shall be uniform at all times. The dumbbell-shaped specimen shall be placed in the grips of the testing machine using care to adjust them symmetrically in order that the tension will be distributed uniformly over the cross section. The tensile stress shall be calculated by dividing the load by the original cross sectional area and shall be expressed in pounds per square inch.

4.6.5 Elongation. Elongation measurements shall be made coincident with the tensile strength tests specified in 4.6.4 to determine compliance with the requirements of 3.5.4. Bench marks located 1 inch apart shall be placed on the 1/2 inch section of each of the three dumbbell-shaped test specimens specified in 4.6.4. The distance between the two bench marks shall be recorded continuously as the specimen is loaded. The testing machine and rate of travel of the jaws shall be the same as specified in 4.6.4.

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TENSILE, ELONGATION TEST SPECIMENS



DIMENSIONS ARE IN INCHES
TOLERANCES:
FRACTIONS 1/32
ANGLES α

TEAR TEST SPECIMEN

FIGURE 1. Test Specimens

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The ultimate elongation shall be calculated by subtracting the original distance between bench marks from the total distance between bench marks at the time of rupture and expressing this difference as a percentage of the original distance.

4.6.6 Compression set. Three specimens, with dimensions of 1 by 2 by 2 inches, shall be cut from three different seat cushion inserts and tested to determine. compliance with the requirements of 3.5.5. The entire test procedure shall be conducted under the following conditions. All gaging, conditioning, and recovery shall be conducted at 73 +2°F and in an atmosphere having a relative humidity of 50 ± 2 percent. After cutting, the specimens shall be conditioned for 3 hours and then placed in an apparatus which will deflect the specimen to 50 ± 1 percent of its height. Within 15 minutes after deflection, the deflected specimen shall be placed in a mechanically convected air oven for a period of 22 hours at a temperature of 158 +2°F and a relative humidity of 5 +1 percent, After removal from the oven, the specimen shall be removed from the deflecting apparatus and allowed to recover for 35 +5 minutes. After recovery, the specimen shall be gaged, and the compression set shall be calculated by expressing the difference between the original height after removal from the apparatus as a percentage of the original height.

4.6.7 Air oven aging. Three seat cushion inserts shall be subjected to an air oven aging test to determine compliance with the requirements of 3.5.6. The inserts shall be placed in a mechanically convected air oven at a temperature of 212° ±2°F for not less than 3 hours. Load deflection characteristics shall then be obtained by the procedure described in 4.6.2.1. The same specimens used in the load deflection charecteristics tests may be used for this test.

4.6.8 Humidity aging. Three specimens, with dimensions of 2 by 2 by 1 inch and cut from three different seat cushion inserts, and three seat cushion inserts shall be subjected to a humidity aging test to determine compliance with the requirements of 3.5.7.1 and 3.5.7.2. The specimens shall be initially dryed for 3 hours at 158° ±2°F in a mechanically convected dry air oven. After drying, the specimens shall be allowed to come to equilibrium after removal from the oven for at least 2 and no more than 24 hours under ambient temperature and humidity conditions as specified in 4.3.1. The humidity test shall be carried out in an autoclave which has been filled to a level of 2 inches above the bottom with freshly distilled water. Both sets of test specimens shall be placed in the autoclave so they do not touch one another or any metal except the supporting surface. With the safety valve open, the water in the autoclave shall be brought to a boil. When all the air is expelled, the safety valve shall be closed and the pressure maintained at 1 to 4 psig (216° to 225°F) for 3 hours. At the end of this time, the specimens shall be removed and dried for 3 hours at 158 ± 2°F in a mechanically convected dry air oven. After removal from the

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oven, the specimens shall be allowed to come to equilibrium for at least 2 hours and not more than 24 hours at ambient conditions as specified in 4.3.1.

4.6.8.1 The load deflection characteristics of the three larger specimens shall be determined as specified in 4.6.2.1. The load deflection losses shall be calculated as follows for six 50 square inch indenter foot values for each complete insert.

$$\text{Load deflection loss} = \frac{Lo - L_1}{L_o} \times 100$$

Where Lo = original load deflection value

and L_1 = load deflection value after humidity test.

4.6.8.2 The three 2 by 2 by 1 inch specimens shall be tested for compression set in accordance with 4.6.6.

4.6.9 Tear resistance. Three specimens as shown on figure 2 shall be cut from three different seat cushion inserts and tested to determine compliance with the requirements of 3.5.8. The tabs of the specimens shall be placed in jaws of a power-driven loading device which will remain at the point of maximum load after rupture. The rate of travel of the jaws shall be 2 inches per minute and uniform at all times. As the specimen is torn, the tear shall be aided with a razor to keep the tear in the center of the specimen. After rupture of the specimen or after at least a 2 inch length is torn, the maximum tearing load shall be noted. The resistance to tear shall be calculated from the maximum load registered and the average height of the specimen. The tear resistance shall be expressed in pounds/inches.

4.7 Inspection of preparation for delivery requirements. Sample items or packs and the inspection of the preservation, packaging, packing, and marking for shipment and storage shall be in accordance with the requirements of section 5, or the documents referenced therein.

5. PACKAGING

5.1 Preservation and packaging shall be level A or C or Standard Practice for Commercial Packaging (see 6.2).

5.1.1.1 Level A. Seat cushion inserts shall be packaged in accordance with method III of MIL-P-116.

5.1.2 Level C. The level C preservation and packaging for seat cushion inserts shall conform to the MIL-STD-794 requirements of this level.

Commercial packaging shall be in accordance with ASTM D 3951-82.

5.2 Packing. Packing shall be level A, B, C or commercial (see 6.2),

5.2.1 Level A. The seat cushion inserts preserved and packaged as specified in 5.1.1.1 shall be packed in a container conforming to PPP-B-601 overseas type. The closure of shipping container shall be in accordance with the appendix of the shipping container specification.

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5.2.2 Level B. The seat cushion inserts shall be preserved and packaged as specified in 5.1.1.1 and shall be packed in a container conforming to PPP-B-636, weather resistant. The closure of shipping container shall be in accordance with the appendix of the specification.

5.2.3 Level C. The seat cushion inserts preserved and packaged as specified in 5.1.1.1 shall be packed in a manner to insure carrier acceptance and safe delivery at destination. Container shall be in accordance with either the uniform freight classification rules or regulations of other carriers, as applicable to the mode of transportation.

or

5.2.3 Commercial packing. Packing shall be accomplished in accordance with ASTM D 3951-82.

5.3 Marking. Interior and exterior containers shall be marked in accordance with the requirements of MIL-STD-129 with the following specific information.

SEAT CUSHION INSERT, POLYURETHANE FOAM, PLASTIC
Applicable Drawing No. _____

6. NOTES

6.1 Intended use. The seat cushion inserts covered by this specification are intended for use in fighter, bomber, and cargo aircraft seat cushions.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification.
b. Application (drawing or specification number).
c. First article requirements.
d. Selection of applicable levels of preservation, packaging, and packing.

6.3 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue,

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-S-27332B(USAF)	SEAT CUSHION INSERT, POLYURETHANE FOAM, PLASTIC, GENERAL

3a. NAME OF SUBMITTING ORGANIZATION SPECIFICATION FOR	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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